PIEZO INSULIN PUMP v2 — REV B DIGITAL HANDOVER PROTOTYPE
BENCH-ONLY: water or 0.9 % saline. NEVER connect to a person or an animal. No real insulin.
NOT FABRICATION-APPROVED. NOT VALIDATED. Contains TBD and PLACEHOLDER parts (Status field / BOM).
Clean ERC/DRC does not prove electrical correctness. Rev A bench validation has not been performed.
Design basis: handoff documents 10 Oct 2026; locked decisions K-1..K-18 preserved. See docs/handover/.

H901
M3

H902
M3

H903
M3

H904
M3

Power input & charger

Battery input, fuse, master switch, BQ25188 charger

File: power_input.kicad_sch

3V3 buck

TPS62840 3.3 V logic rail

File: power_3v3.kicad_sch

48V flyback

LM51551 non-isolated flyback, 48 V motor rail

File: power_48v.kicad_sch

Kill path (fail-off)

48 V enable kill path: main MCU enable, supervisor veto

File: safety.kicad_sch

Main MCU STM32WBA55

Main MCU STM32WBA55CG: dosing, BLE, encoder, PMD401 SPI

File: mcu_main.kicad_sch

Supervisor STM32C031

Supervisor STM32C031C6: heartbeat, park line, 48 V veto, own encoder count

File: mcu_supervisor.kicad_sch

PMD401 & encoder interface

PMD401 servo/status/power/RS-485 connectors, LL06 encoder, protection

File: pmd_interface.kicad_sch

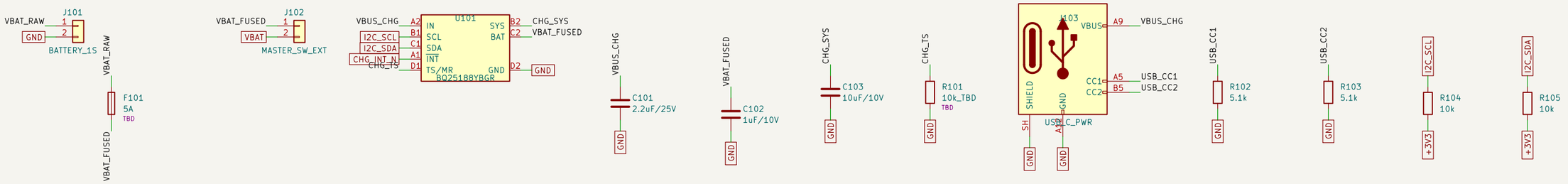
Root: hierarchy, mounting, disclaimers
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts — see Status field and docs/handove
Piezo Insulin Pump v2 — BENCH ONLY (water/saline), never on a person or animal

Sheet: /
File: pump_v2.kicad_sch

Title: Piezo Insulin Pump v2 — Rev B handover prototype

Size: A3	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 1/8

Battery path (E-01 DECIDED): 1S protected cell -> J101 -> F101 5 A fuse -> J102 master switch -> VBAT.
Charger (K-12 BQ25188) charges the cell on VBAT_FUSED. E-09: the charger is NEVER the load path:
BQ25188 SYS (CHG_SYS) has only its required 10 uF and no load. Charge current set over I2C (OPEN B.19).
Cell protection is inside the pack (K-12 '1S protected cell'). Cell size/connector OPEN B.20.



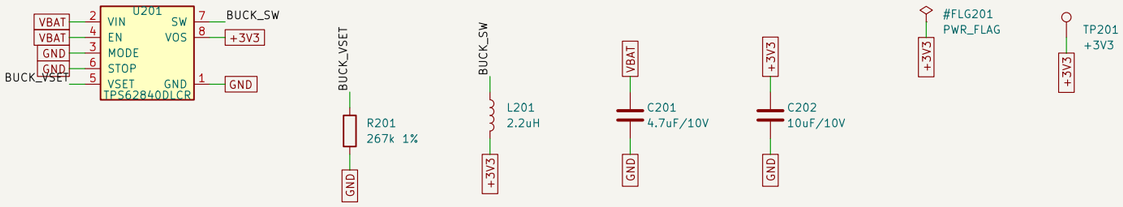
Sheet 2: Power input & charger
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts - see Status field and docs/handover
Piezo Insulin Pump v2 - BENCH ONLY (water/saline), never on a person or animal

Sheet: /Power input & charger/
File: power_input.kicad_sch

Title: Battery input, fuse, master switch, BQ25188 charger

Size: A3	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 2/8

K-12: TPS62840 for 3.3 V on Rev B. Taken from VBAT (not from the 48 V chain), as decided for Rev A.
Below about 3.4 V in, the buck runs in 100 % mode and 3V3 follows VBAT (E-02 >=3.0 V while dosing; S-13 3.4 V stop).
Note CF-8: E-02 wording says 'LDO' (Rev A); Rev B uses the decided TPS62840 buck.



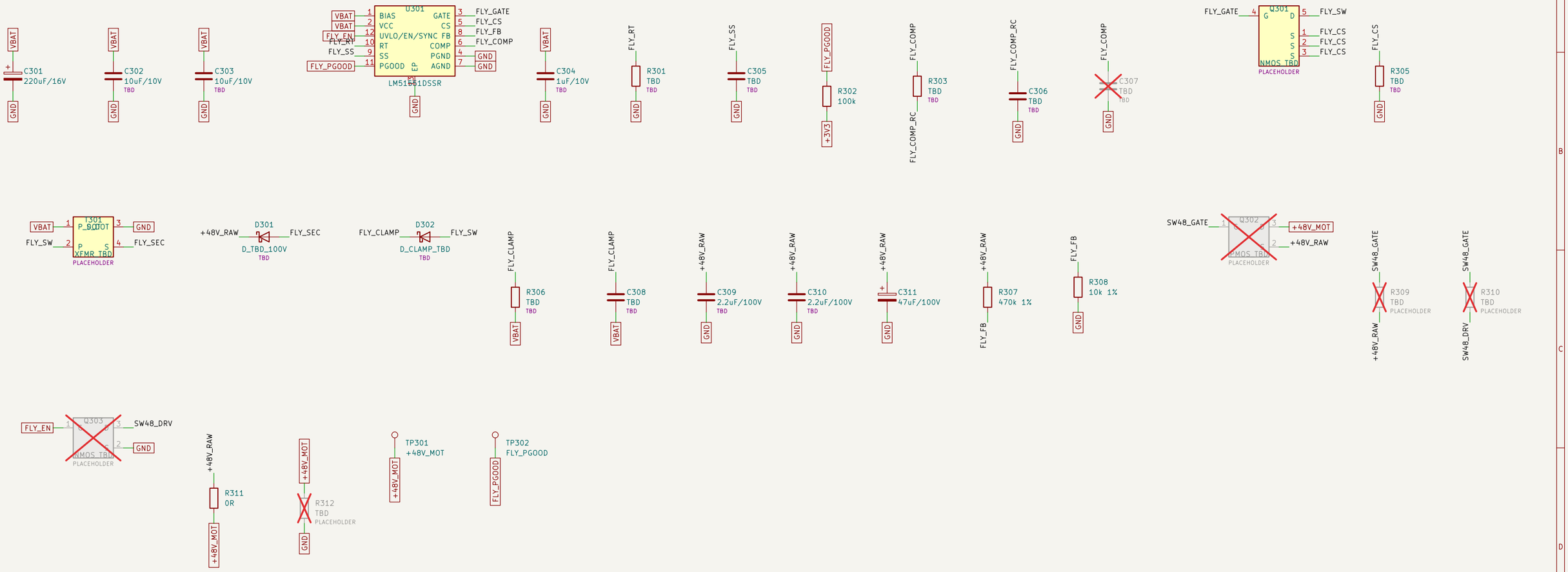
Sheet 3: 3V3 buck
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts — see Status field and docs/handove
Piezo Insulin Pump v2 — BENCH ONLY (water/saline), never on a person or animal

Sheet: /3V3 buck/
File: power_3v3.kicad_sch

Title: TPS62840 3.3 V logic rail

Size: A3	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 3/8

K-12 / DECIDED 29 Sep: Rev B 48 V stage is an LM51551 flyback (plain boost hits the 93 % duty limit).
Topology follows TI SNVSB75E Figure 10-12 'Typical Non-Isolated Flyback'; BIAS tied to VCC for 1S start (>=2.97 V).
POWER STAGE NOT DESIGNED: transformer, FET, diode, sense resistor, clamp, RT, SS and compensation are TBD (OPEN B.16).
Kill path: UVLO/EN driven by FLY_EN (sheet 'Kill path'). EN < 0.4 V min (SNVSB75E) = shutdown. A flyback has no DC path
from VBAT to the output, so EN low de-energises the rail. K-12 'switched output' + E-11 bleeder: DNP PLACEHOLDERS only.
E-06: every 48 V capacitor rated >= 100 V. E-03: 45.6-50.4 V at the PMD401.



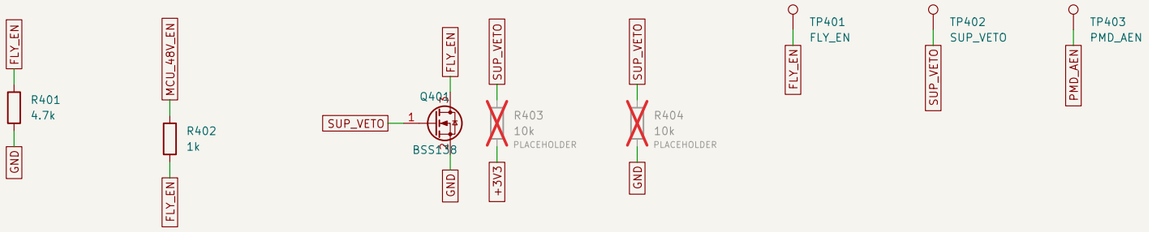
Sheet 4: 48V flyback
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts — see Status field and docs/handover
Piezo Insulin Pump v2 — BENCH ONLY (water/saline), never on a person or animal

Sheet: /48V flyback/
File: power_48v.kicad_sch

Title: LM51551 non-isolated flyback, 48 V motor rail

Size: A3	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 4/8

K-10 DECIDED (fail-off): 4.7 k pull-down on the enable, main MCU drives it through 1 k, supervisor veto via one BSS138.
On Rev B the enable node is LM51551 UVLO/EN (FLY_EN). EN rising threshold 0.4-0.7 V, UVLO 1.425-1.575 V (SNVSB75E).
MCU high: 3.3 V x 4.7/(4.7+1) = 2.72 V -> run. Pull-down only: 0 V -> shutdown. BSS138 on: ~0 V -> shutdown.
UNRESOLVED (OPEN item 3 / CF-9): BSS138 gate default. R403 (pull-UP, veto on by default) and R404 (pull-DOWN,
as wired on Rev A) are BOTH placed DNP. Owner decision required before build. With neither fitted the gate floats
while the supervisor is in reset - this is NOT an acceptable build state. SG-03 must be run after the decision.



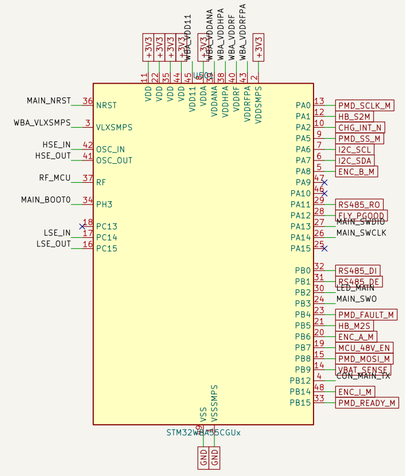
Sheet 5: Kill path (fail-off)
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts - see Status field and docs/handover
Piezo Insulin Pump v2 - BENCH ONLY (water/saline), never on a person or animal

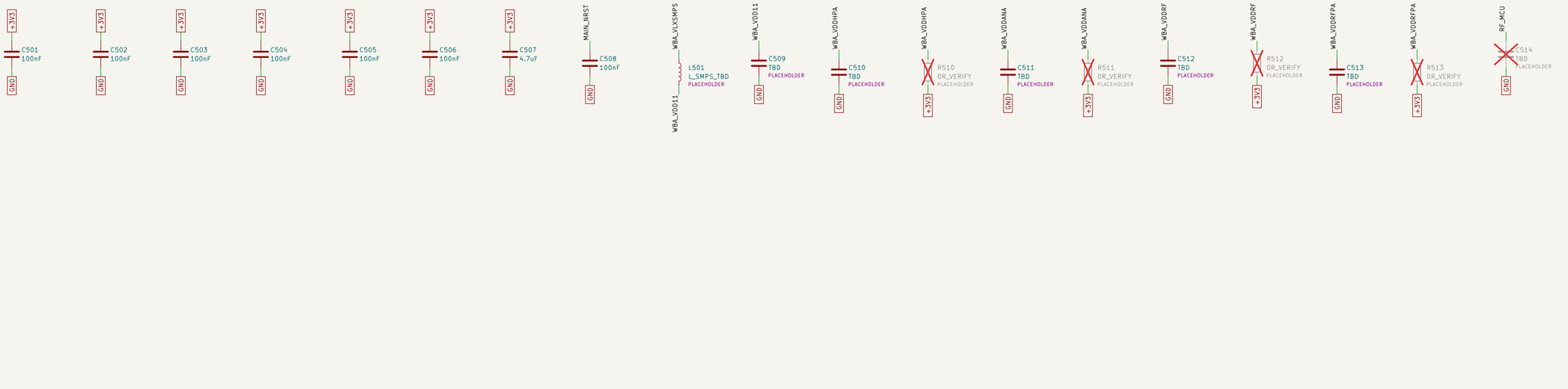
Sheet: /Kill path (fail-off)/
File: safety.kicad_sch

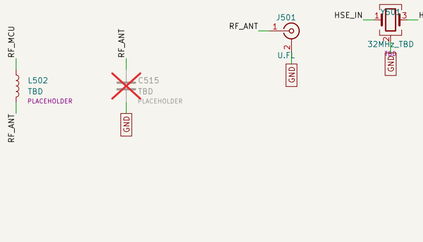
Title: 48 V enable kill path: main MCU enable, supervisor veto

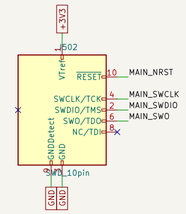
Size: A3	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 5/8

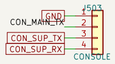
Pin map DERIVED from ST pin data in the KiCad STM32WBA55CGUx symbol: CONFIRM IN STM32CubeMX before layout freeze.
Encoder on TIM2_CH1/CH2 (PB6/PA8) — TIM2 width must be confirmed in RM0493 (CF-21/FW-08).
SPI3 -> PMD401 (mode 3, <= 500 kbit/s, PMD401 manual 150025-01 sec.5.2). USART2 + DE -> MAX3485. LPUART1 <-> supervisor link.
USART1 TX-only console (PA8, the only USART1_RX pin, carries the encoder).
ST DATASHEET NOT AVAILABLE in this session (st.com unreachable): SMPS inductor, VDD11, VDDRF/VDDRFPA/VDDHPA/VDDANA
connections, decoupling values, HSE crystal/load caps, BOOT0 strap and RF matching are PLACEHOLDERS (see ERC report).









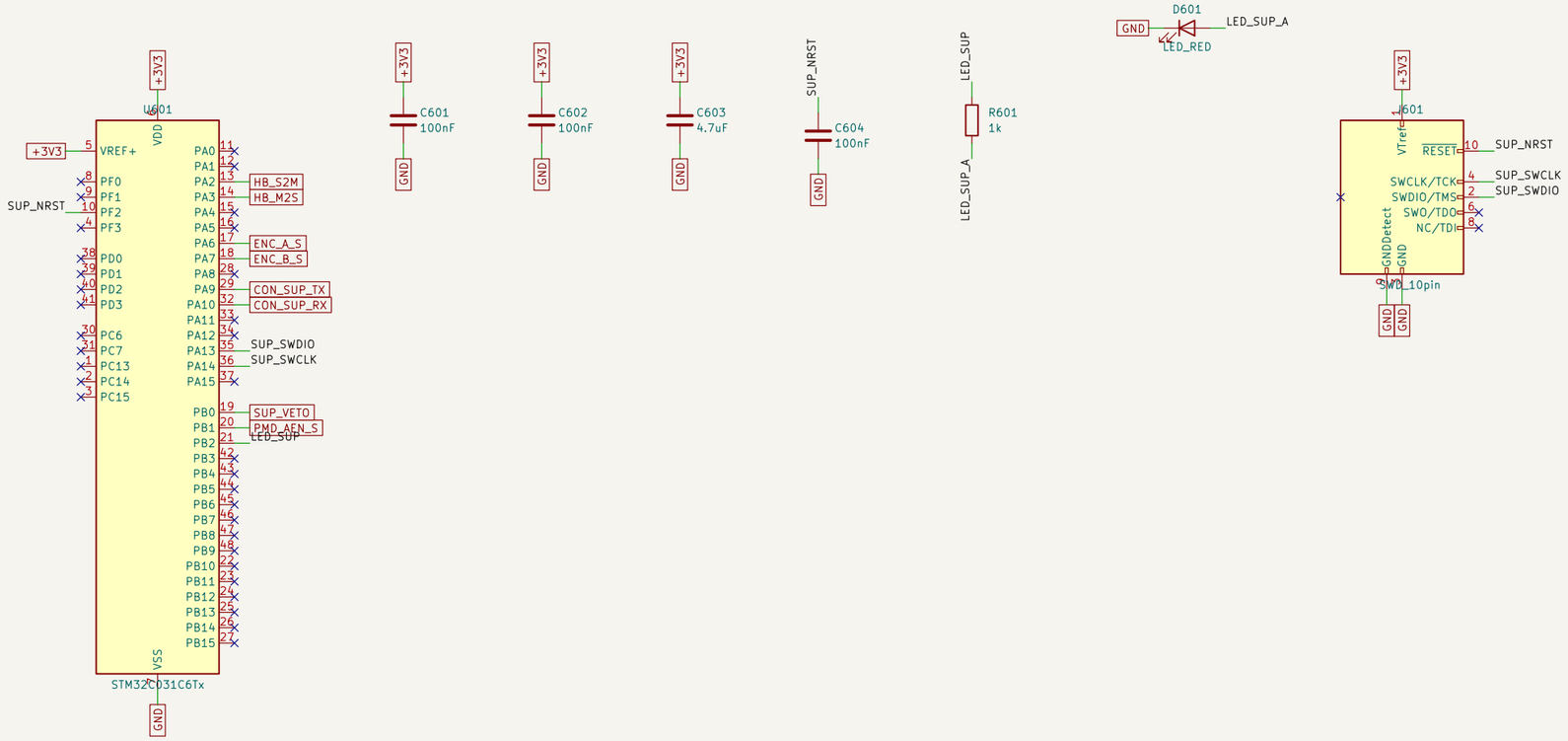


Sheet 6: Main MCU STM32WBA55
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts — see Status field and docs/handover
Piezo Insulin Pump v2 — BENCH ONLY (water/saline), never on a person or animal
Sheet: /Main MCU STM32WBA55/
File: mcu_main.kicad_sch

Title: Main MCU STM32WBA55CG: dosing, BLE, encoder, PMD401 SPI

Size: A2	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 6/8

K-6 / F-09: separate MCU, separate code base. It can only STOP: veto on the 48 V enable (SUP_VETO) and park line (PMD_AEN).
S-07 (approved 10 Oct): encoder A/B also into TIM3 CH1/CH2 via its own series resistors.
Pin map DERIVED from the KiCad STM32C031C6Tx symbol (ST pin data): CONFIRM in CubeMX. NRST assumed on PF2 (PF2-NRST) - VERIFY.
Package LQFP48 (C6T6) ASSUMED; the C031C6 also exists in UFQFPN48 (C6U6).



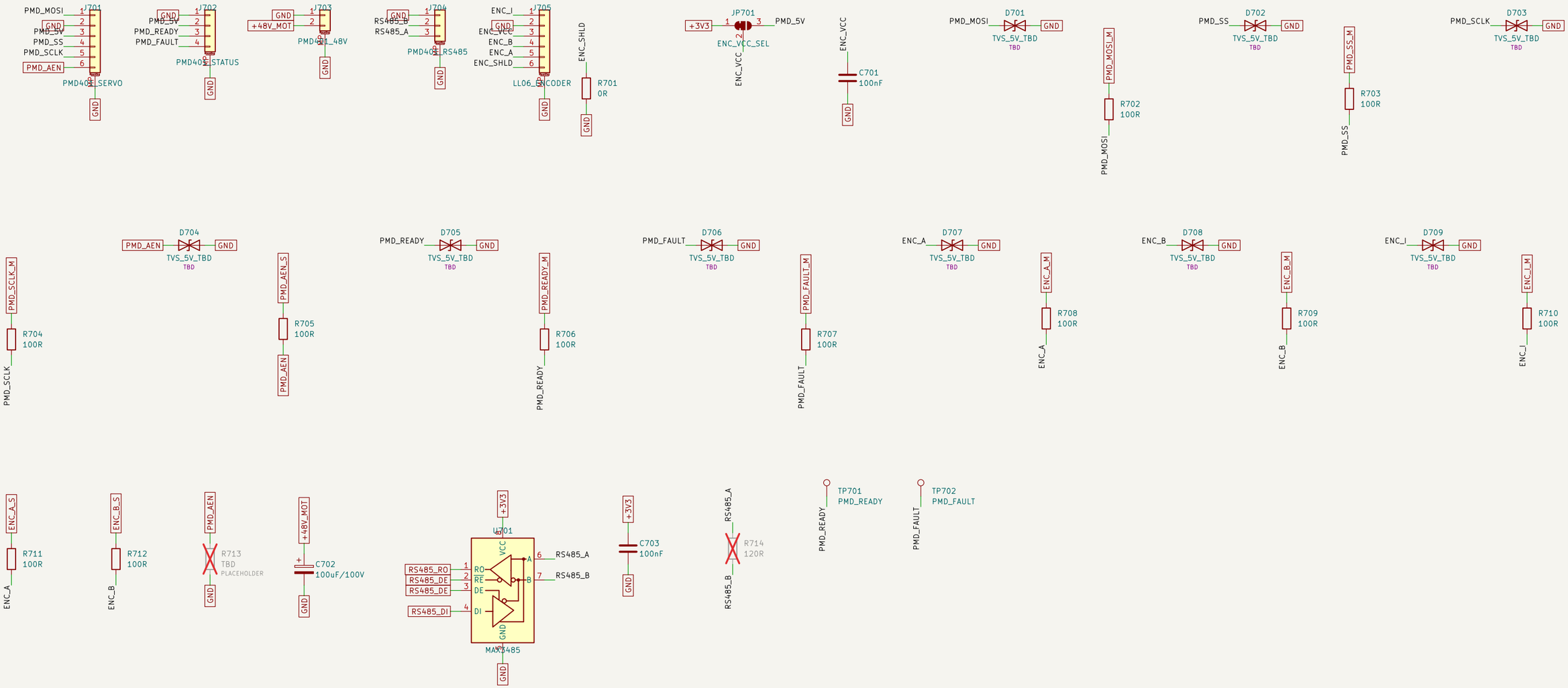
Sheet 7: Supervisor STM32C031
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts - see Status field and docs/handove
Piezo Insulin Pump v2 - BENCH ONLY (water/saline), never on a person or animal

Sheet: /Supervisor STM32C031/
File: mcu_supervisor.kicad_sch

Title: Supervisor STM32C031C6: heartbeat, park line, 48 V veto, own en

Size: A3	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 7/8

Connector pinouts CONFIRMED from PiezoMotor PMD401 Technical Manual 150025-01 sec.3.4 / 3.5 (servo mode). All JST SH 1.0 mm.
Servo connector = the PMD401 6-pole sensor connector in servo mode; the LL06 encoder therefore connects to THIS board (J705).
J705 mirrors the PMD401 sensor-connector pinout so the kit encoder cable can plug in - VERIFY against the delivered cable.
E-07: TVS + 100 R on every PMD401 signal line. PMD401 inputs have internal pull-ups (10k/~20k/22k) to an UNSTATED rail (IF-02).
AEN has a 22k internal pull-up (manual): a dead supervisor leaves the motor ENABLED unless R706 is fitted (OPEN item 4).
Encoder supply (OPEN item 12): JP701 default 3V3 (logic rail, survives 48 V switching); alternative PMD401 5V Out.



Sheet 8: PMD401 & encoder interface
ERC/DRC clean status does not prove electrical correctness
NOT fabrication-approved. Contains TBD / PLACEHOLDER parts - see Status field and docs/handove
Piezo Insulin Pump v2 - BENCH ONLY (water/saline), never on a person or animal

Sheet: /PMD401 & encoder interface/
File: pmd_interface.kicad_sch

Title: PMD401 servo/status/power/RS-485 connectors, LL06 encoder, pro

Size: A3	Date: 2026-10-10	**Rev: B-draft-handover**
KiCad E.D.A. 10.0.3		Id: 8/8